RAVICHETHAN RAMAIAH

ASPIRING ELECTRONICS & EMBEDDED SYSTEMS ENGINEER | VLSI | SOFTWARE DEVELOPMENT

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PROFESSIONAL SUMMARY

Aspiring engineer interested in embedded systems, electronics, VLSI design, and software development. Building practical skills through Arduino-based electronics work and programming projects. Familiar with foundational C/C++, Python, web development concepts, Git/GitHub workflows, and introductory System-on-Chip (SoC) design topics. Seeking an entry-level opportunity or internship to keep learning and contribute to hands-on engineering work.

TECHNICAL SKILLS

Programming: C, C++, Python (foundational knowledge)

Web development: HTML, CSS, JavaScript (foundational knowledge)

Electronics & components: Arduino Uno, ultrasonic sensor, servo motor, DC motors, L298N motor driver, HC-05 Bluetooth module, BC547 transistors, LEDs, buzzer, resistors

Embedded systems: Microcontroller fundamentals, sensor interfacing concepts, basic motor-control logic

VLSI fundamentals: SoC architecture, ASIC vs. FPGA, introductory ASIC design flow

Tools: Visual Studio Code, Git, GitHub, browser developer tools

PROJECTS

Student Project Tracker

Web Development | In progress

- Building a web application to organize student projects by name and status.
- Practiced HTML structure, CSS styling, JavaScript event handling, objects, arrays, and form submission.
- Implemented project create/read/update/delete interactions and browser storage during earlier milestones; currently learning Express.js API and server-side project structure.
- Using Git and GitHub to track source-code changes.

Obstacle-Avoiding Robot

Embedded Systems & Robotics | Project work / learning

- Exploring an Arduino Uno robot concept that detects obstacles with an ultrasonic sensor.
- Planned components include a servo motor, L298N motor driver, DC motors, rechargeable batteries, and an HC-05 Bluetooth module.
- Learning how sensor readings can inform motor-control decisions and how Bluetooth can support wireless control.

Water-Level Indicator and Alert Circuit

Electronics | Circuit concept

- Explored a water-level indicator concept using BC547 NPN transistors and multiple LED indicators.
- Included a buzzer concept for an audible alert and studied the role of resistors and transistor switching.

Personal Portfolio Website

Frontend Development

- Created a beginner portfolio page using HTML and CSS.
- Practiced page structure, typography, styling, and layout fundamentals for presenting personal and technical work.

TECHNICAL LEARNING

VLSI System On Chip Design – OVERVIEW

Studied SoC fundamentals, smartphone SoC architecture, ASIC versus FPGA, and an introductory design flow covering specification, RTL, verification, gate-level simulation, DFT, STA, GDSII, and AMS concepts. Confirm certificate/completion details before listing as a formal certification.

Internship on Embedded System Design – 30 Days

Participating in technical learning covering embedded-system fundamentals and related tools/topics, including serial communication concepts (UART, SPI, I^2C), JTAG, and basic debugging instruments. Update status and provider details before applying.

EDUCATION

B-Tech in ECE [2025-2029]

REVA University

CAREER OBJECTIVE

To begin a career or internship in embedded systems, electronics, VLSI, or software development, applying foundational technical knowledge while continuing to develop practical engineering and problem-solving skills.